

**Time:** 5 minutes**Outcome:** Calculate output per worker and interpret productivity changes.**Difficulty:** ●○○ Beginner

Productivity Measurement



THE CORE IDEA

Labor productivity is real output per unit of labor input, often per hour worked. Divide real output by labor hours and keep the measurement period consistent. Higher total output alone does not prove productivity rose; hours may have risen faster.



HOW TO RECOGNIZE IT

- Measure labor productivity as real output per worker or per hour worked.
- Use the same output and labor units when comparing periods.
- An increase in total output need not raise productivity if labor input also rises.



WATCH OUT

Do not confuse total output with output per unit of labor. Productivity compares output with the labor input used to produce it, so changes in both output and labor matter.



WORKED EXAMPLE: MEASURING LABOR PRODUCTIVITY

A workshop produces 2,400 units using 300 labor hours, so productivity is $2,400 / 300 = 8$ units per hour. Next month it produces 2,700 units using the same 300 hours: 9 units per hour. Productivity rises by $(9 - 8) / 8 \times 100 = 12.5\%$. More output per labor hour, not simply more total hours, is the improvement.



CHECK YOURSELF

A business uses more labor hours but produces the same amount of output. What happens to labor productivity?

**READY?**

Return to the game and master this concept.